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M.Sc Computer Science

MASTER THESIS PROPOSAL

Uncertainty-Aware Machine Learning for Microwave-Based Body Composition Assessment

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1 Title

"Uncertainty-Aware Machine Learning for Microwave-Based Body Composition Assessment"

2 Abstract

Noninvasive microwave sensing has emerged as a promising technique for biomedical diagnostics. However, a critical limitation for clinical translation is that current machine learning models typically output point estimates without prediction confidence. In real clinical settings, measurement noise and biological variability can lead to unreliable predictions. This thesis aims to develop and evaluate an uncertainty-aware machine learning system for estimating skin, fat, and muscle properties. By integrating Bayesian techniques and quantifying predictive uncertainty, this work seeks to provide a clinically interpretable risk score to support safer medical decision-making.

3 Background

Noninvasive microwave sensing has emerged as a promising technique for biomedical diagnostics due to its safety, low cost, and ability to probe tissue composition without ionizing radiation. Recent research, including Viktor Mattsson's doctoral thesis on data-driven methods for microwave sensor devices, has demonstrated that machine learning models can estimate tissue-layer properties such as skin, fat, and muscle from microwave measurements [1].

However, a critical limitation remains for clinical translation. Current machine learning models typically output point estimates without providing information about prediction confidence. In real clinical settings, microwave measurements are affected by sensor placement variability, anatomical differences, measurement noise, and imperfect phantom-to-human correspondence. These uncertainties can lead to unreliable predictions if not properly quantified.

Mattsson's thesis repeatedly highlights the role of measurement uncertainty and statistical variability, particularly in bone measurements and craniosynostosis monitoring, where the lack of confidence estimation limited clinical interpretability [1]. This motivates the need for uncertainty-aware ma-

chine learning models that not only predict tissue properties but also quantify how reliable those predictions are.

4 Tasks

The main aim of this thesis is to develop and evaluate an uncertainty-aware machine learning system. The specific objectives are:

4.1 Model Development

Develop regression models that predict tissue-layer thickness or dielectric properties from microwave sensor measurements.

4.2 Uncertainty Integration

Integrate Bayesian uncertainty estimation techniques into the prediction pipeline and quantify predictive uncertainty using established metrics.

4.3 Analysis and Risk Scoring

Analyze how measurement uncertainty propagates from sensor data to final tissue estimates and propose a clinically interpretable risk or confidence score to support medical decision-making.

5 Methods

5.1 Data Sources

The thesis will use existing datasets available from the microwave sensing research group:

- Phantom measurements with known tissue-layer configurations.
- Volunteer or limited clinical datasets where tissue thickness is validated using ultrasound or reference methods.
- Optional simulated datasets for controlled noise and variability studies.

Each sample may include frequency-dependent S-parameters (S11, S21) and metadata related to measurement conditions.

5.2 Baseline Machine Learning Models

Initial models will be developed as deterministic baselines to serve as references:

- Random Forest Regression
- XGBoost Regression
- Fully Connected Neural Networks

5.3 Uncertainty-Aware Modeling Approaches

The thesis will investigate and compare multiple uncertainty estimation techniques:

- **Bayesian Neural Networks:** Treat network weights as probability distributions and approximate posterior inference using variational methods.
- **Monte Carlo Dropout:** Apply dropout during inference to approximate Bayesian uncertainty by performing multiple stochastic forward passes.
- **Deep Ensembles:** Train multiple independent models with different initializations to estimate uncertainty from prediction diversity.

5.4 Uncertainty Calibration and Evaluation

To ensure meaningful uncertainty estimates, calibration will be assessed using Expected Calibration Error (ECE), Reliability diagrams, Prediction interval coverage probability, and Sharpness versus calibration trade-off analysis.

5.5 Uncertainty Propagation and Risk Score

The thesis will analyze how uncertainty arises from sensor noise, frequency-domain variability, and domain shift. Based on these estimates, a risk or confidence score will be developed to flag predictions with high uncertainty, allowing clinicians to distinguish reliable from unreliable estimates.

6 Evaluation Metrics

The models will be evaluated using the following metrics:

- Mean Absolute Error (MAE) and Root Mean Square Error (RMSE).
- Prediction interval coverage.
- Calibration metrics (ECE, reliability curves).
- Sensitivity to noise and perturbations.

7 Delimitations

- **Clinical Translation Gap:** Current conventional ML models do not indicate when predictions are unreliable, which is a barrier to clinical adoption.
- **Data Variability:** Measurement noise, sensor placement variability, and anatomical differences are not explicitly modeled in standard deterministic approaches.

8 Time Plan

- **Weeks 1–3:** Literature review and dataset familiarization.
- **Weeks 4–6:** Baseline ML model development (Random Forest, XGBoost, etc.).
- **Weeks 7–11:** Bayesian and uncertainty-aware model implementation (BNNs, MC Dropout, Ensembles).

- **Weeks 12–14:** Calibration and uncertainty evaluation.
- **Weeks 15–17:** Risk score development and uncertainty propagation analysis.
- **Weeks 18–20:** Thesis writing, revision, and presentation preparation.

References

[1] V. Mattsson, "Data-driven methods for microwave sensor devices in musculoskeletal diagnostics," Ph.D. dissertation, Uppsala University, 2024, acta Universitatis Upsaliensis. [Online]. Available: https://urn.kb.se/resolve?urn=urn:nbn:se:uu:diva-521537

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